

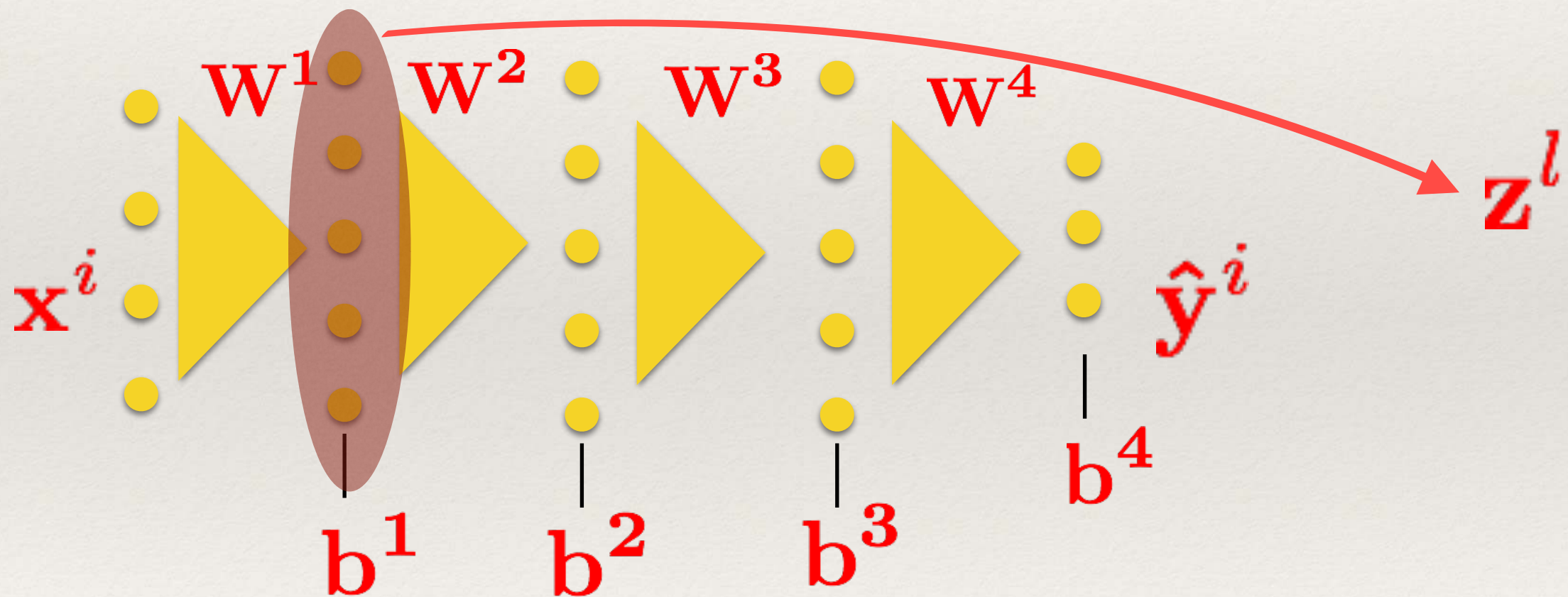
E9 205 Machine Learning for Signal Processing

Deep Learning

23-11-2016

Deep Feed Forward Networks

Neural networks with multiple hidden layers - Deep networks



Back Propagation Algorithm

Variables

$\mathbf{x}^i$ Input of DNN

$\hat{\mathbf{y}}^i$ Output of DNN

$\mathbf{y}^i$ Desired Output

$\mathbf{W}_t^l$ $\mathbf{b}_t^l$ Weights layer l iteration t

m_b Mini-batch size

$\mathbf{z}_t^l$ Hidden layer output at l iteration t

$\mathbf{h}_t^l$ Hidden layer output at l iteration t

J Cost Function

$g(\cdot)$ Hidden Activations

Gradient Descent Algorithm

$$\mathbf{W}_{t+1}^l = \mathbf{W}_t^l - \epsilon \Delta \mathbf{W}_t^l$$

$$\mathbf{b}_{t+1}^l = \mathbf{b}_t^l - \epsilon \Delta \mathbf{b}_t^l$$

Define $\mathbf{e}_t^l = \frac{\partial J}{\partial \mathbf{h}_t^l}$

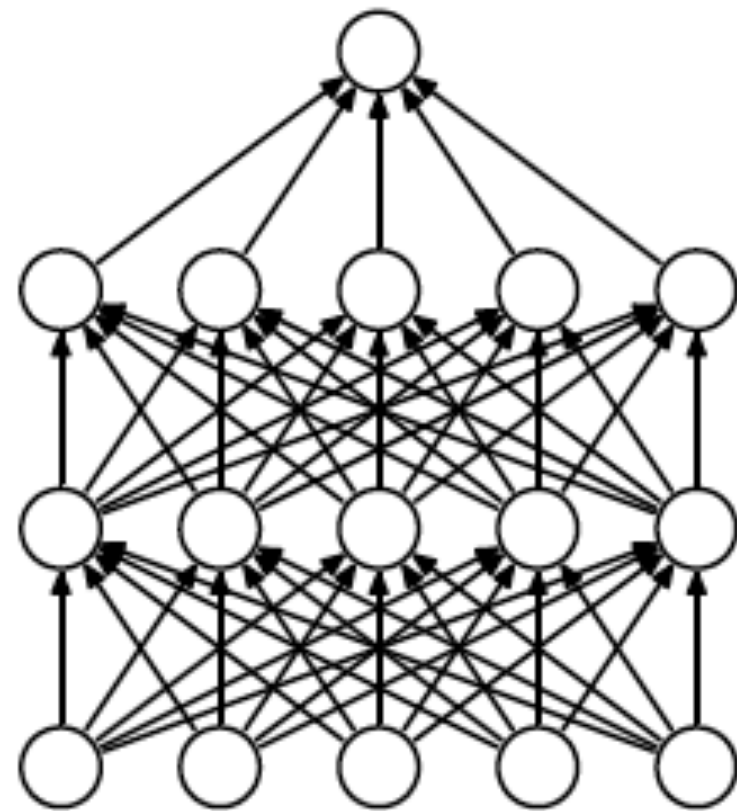
Then, $\mathbf{e}_t^{l-1} = (\mathbf{W}_t^l)^T \{g'(\mathbf{z}_t^l) \cdot \mathbf{e}_t^l\}$

$$\Delta \mathbf{W}_t^l J = \{g'(\mathbf{z}_t^l) \cdot \mathbf{e}_t^l\} (\mathbf{h}_t^{l-1})^T$$

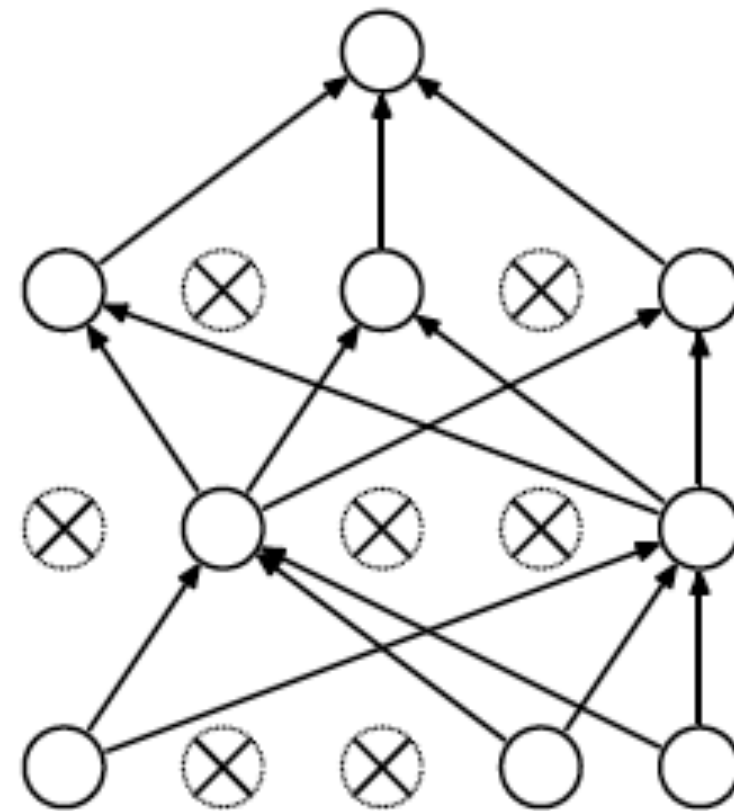
$$\Delta \mathbf{b}_t^l J = \{g'(\mathbf{z}_t^l) \cdot \mathbf{e}_t^l\}$$

- Element wise product

Regularization - Dropout

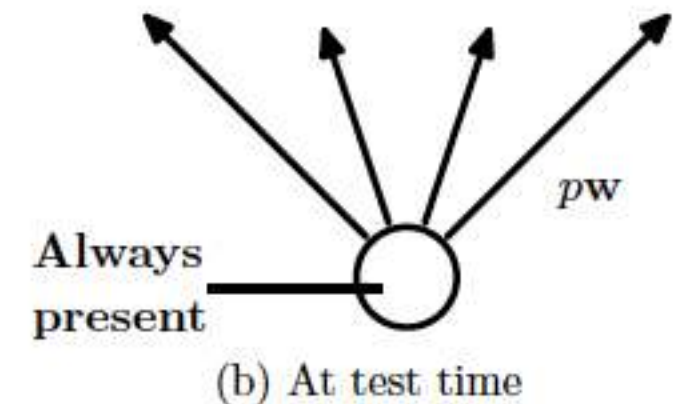
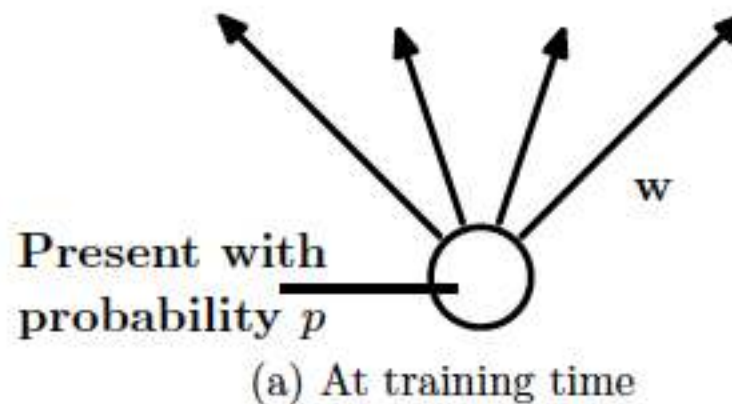


(a) Standard Neural Net

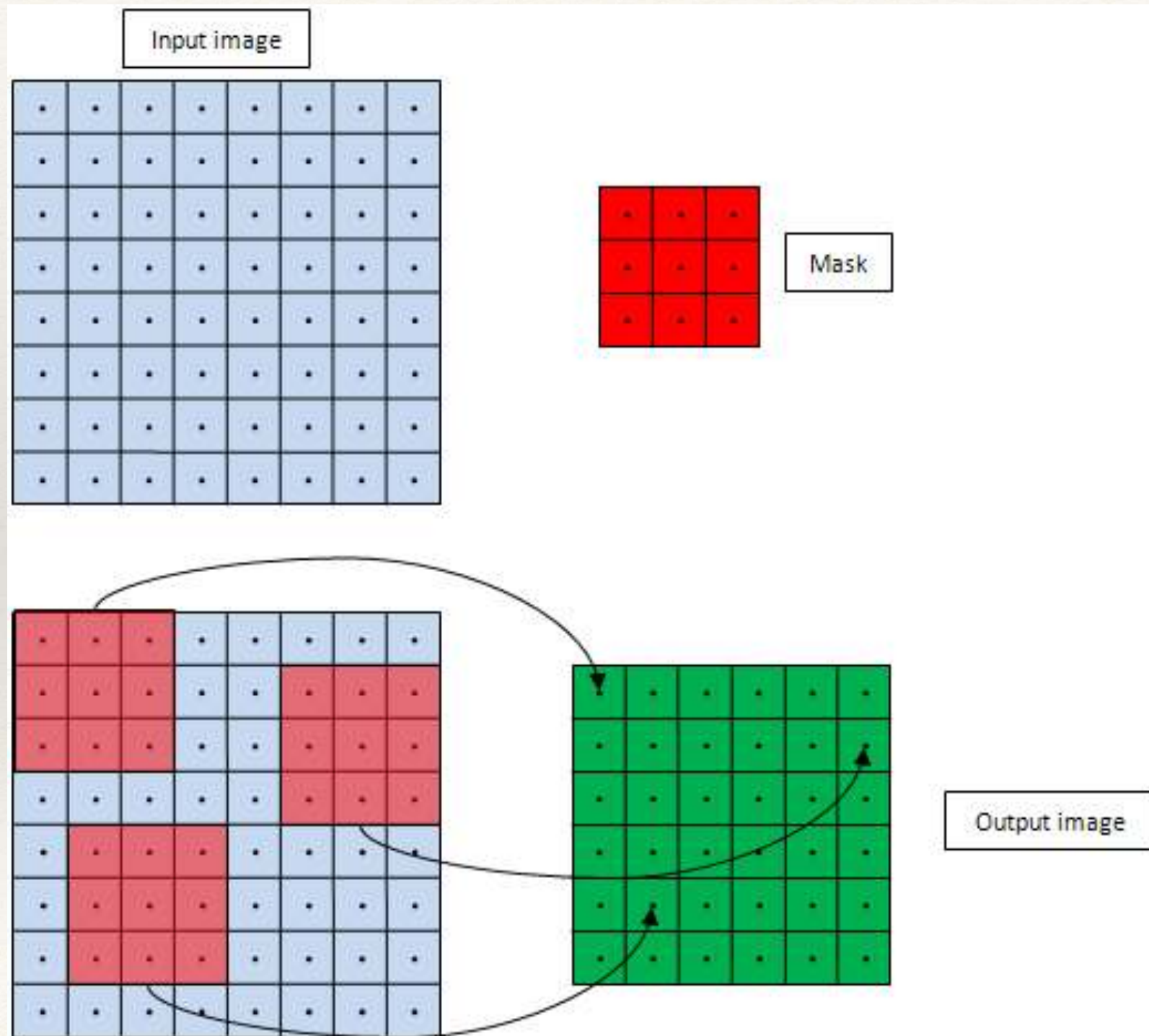


(b) After applying dropout.

[Srivatsava, 2014]



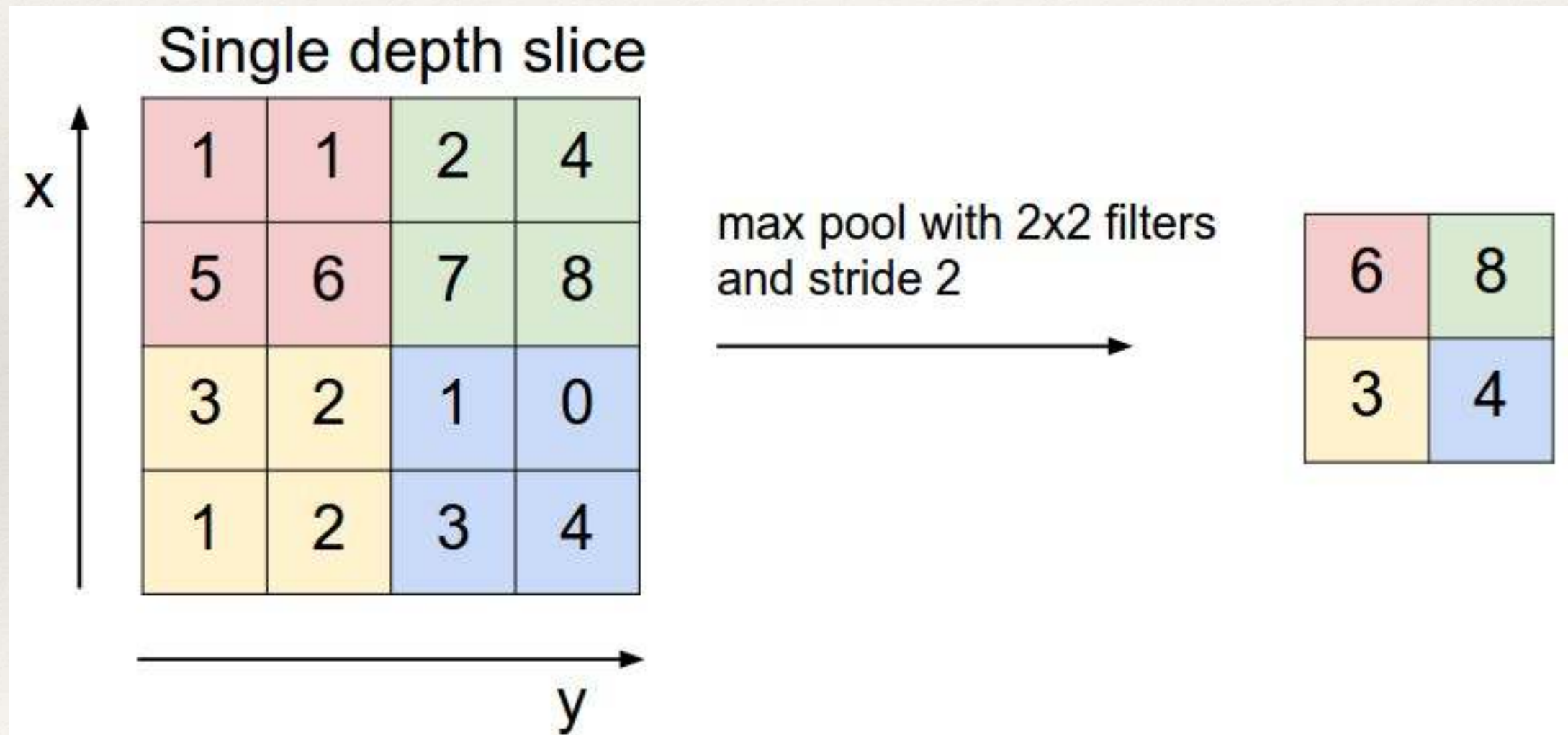
Convolutional Network



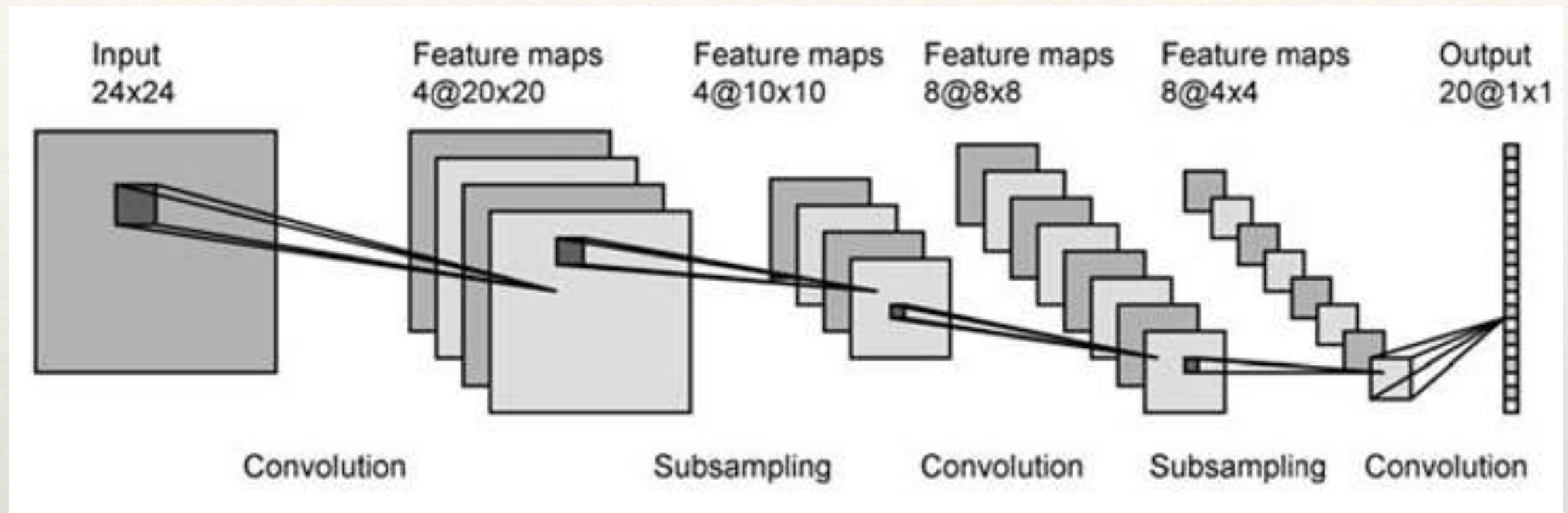
Weight sharing

Convolutional Neural Network

Max Pooling

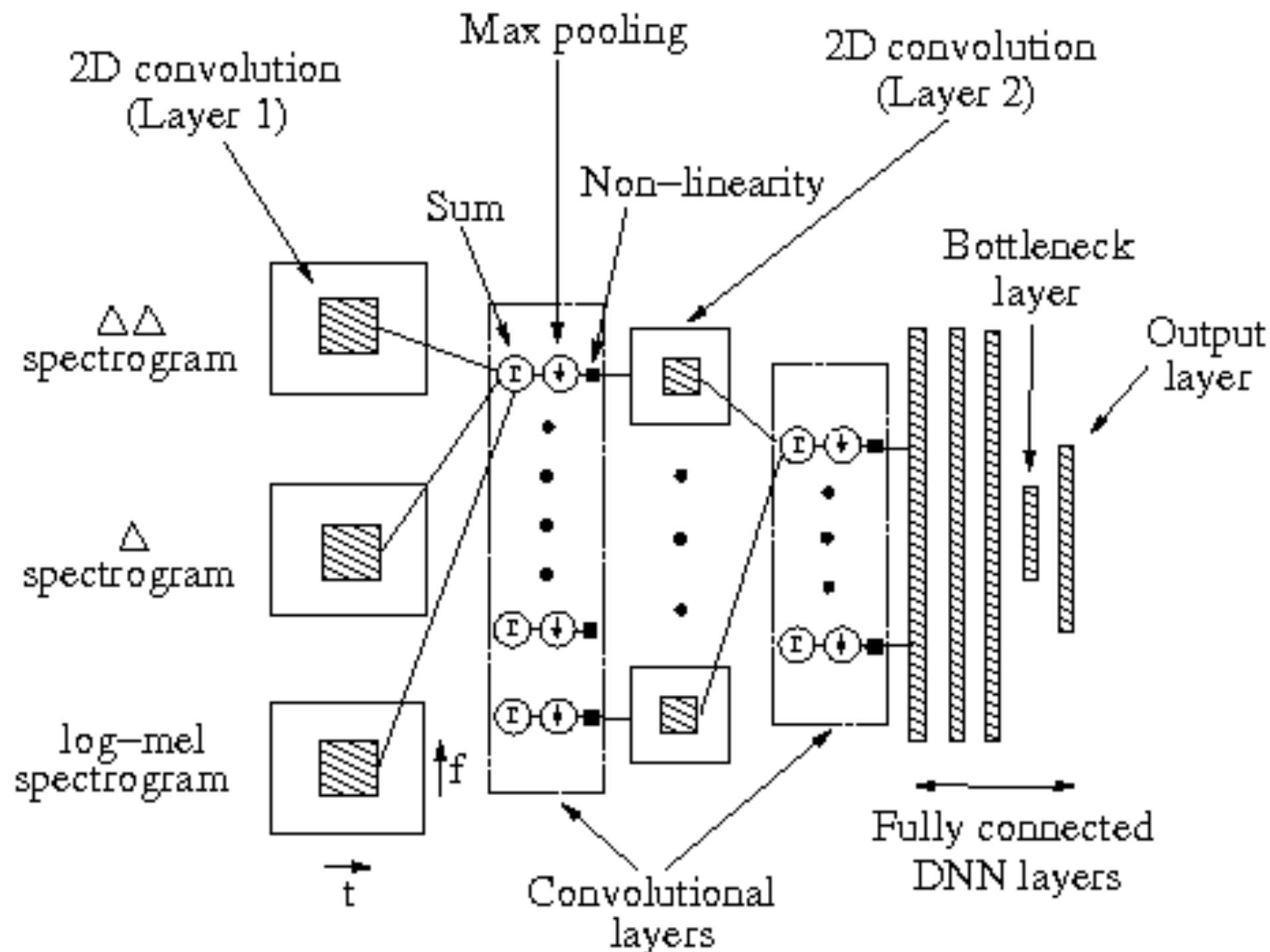


Convolutional Neural Network

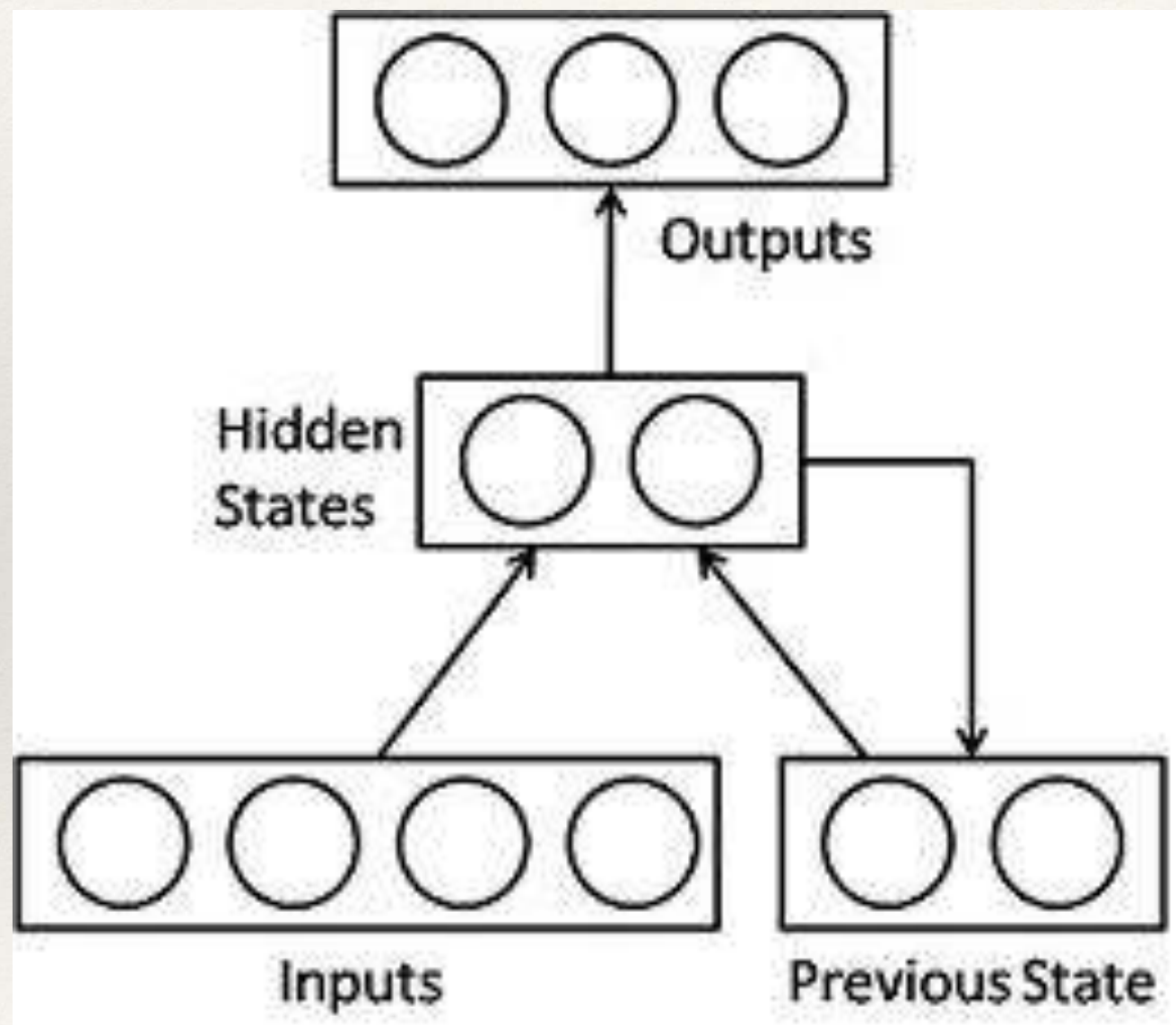


- Successive layers of convolutions and subsampling operation [LeCun, 98].
- **Non-linearities (ReLu or Sigmoid)** are typically applied after each subsampling stage.

CNNs in Speech Processing



Recurrent Networks

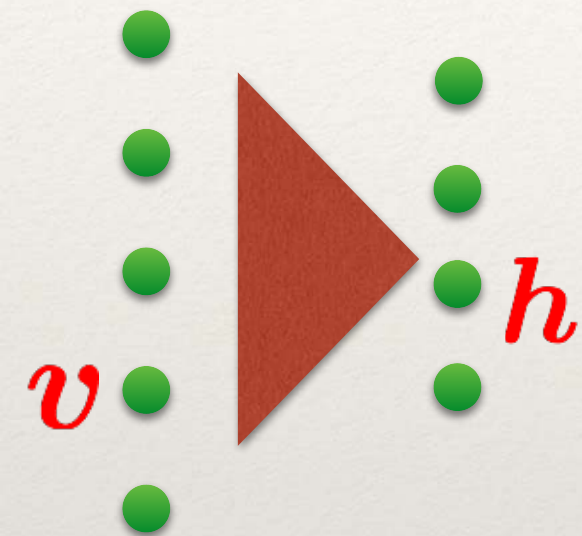


Feedback

Restricted Boltzmann Machines

Initializing large networks with **deep belief**

networks (DBN)



- Gaussian Bernoulli RBM - Gaussian visible layer and Bernoulli hidden layer.
- Define an energy function

$$E(\mathbf{v}, \mathbf{h}) = 0.5(\mathbf{v} - \mathbf{a})^T (\mathbf{v} - \mathbf{a}) - \mathbf{b}^T \mathbf{h} - \mathbf{h}^T \mathbf{W} \mathbf{v}$$

- Define joint probability density $P(\mathbf{v}, \mathbf{h}) = \frac{e^{-E(\mathbf{v}, \mathbf{h})}}{Z}$
- Estimate the parameters $\mathbf{a}$, $\mathbf{b}$, $\mathbf{W}$ by maximizing the joint density

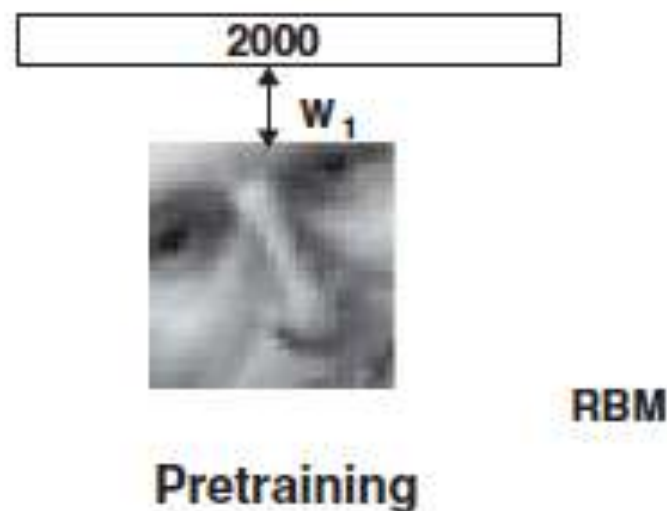
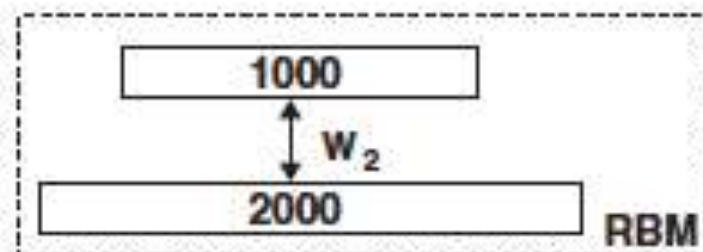
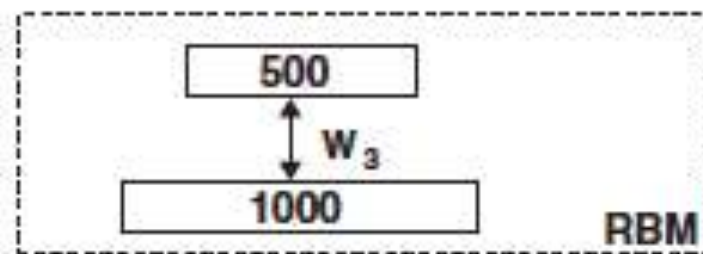
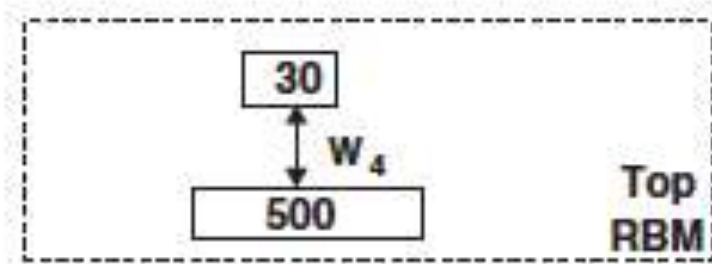
Restricted Boltzmann Machines

It can be shown that probability of hidden unit being is the sigmoid function

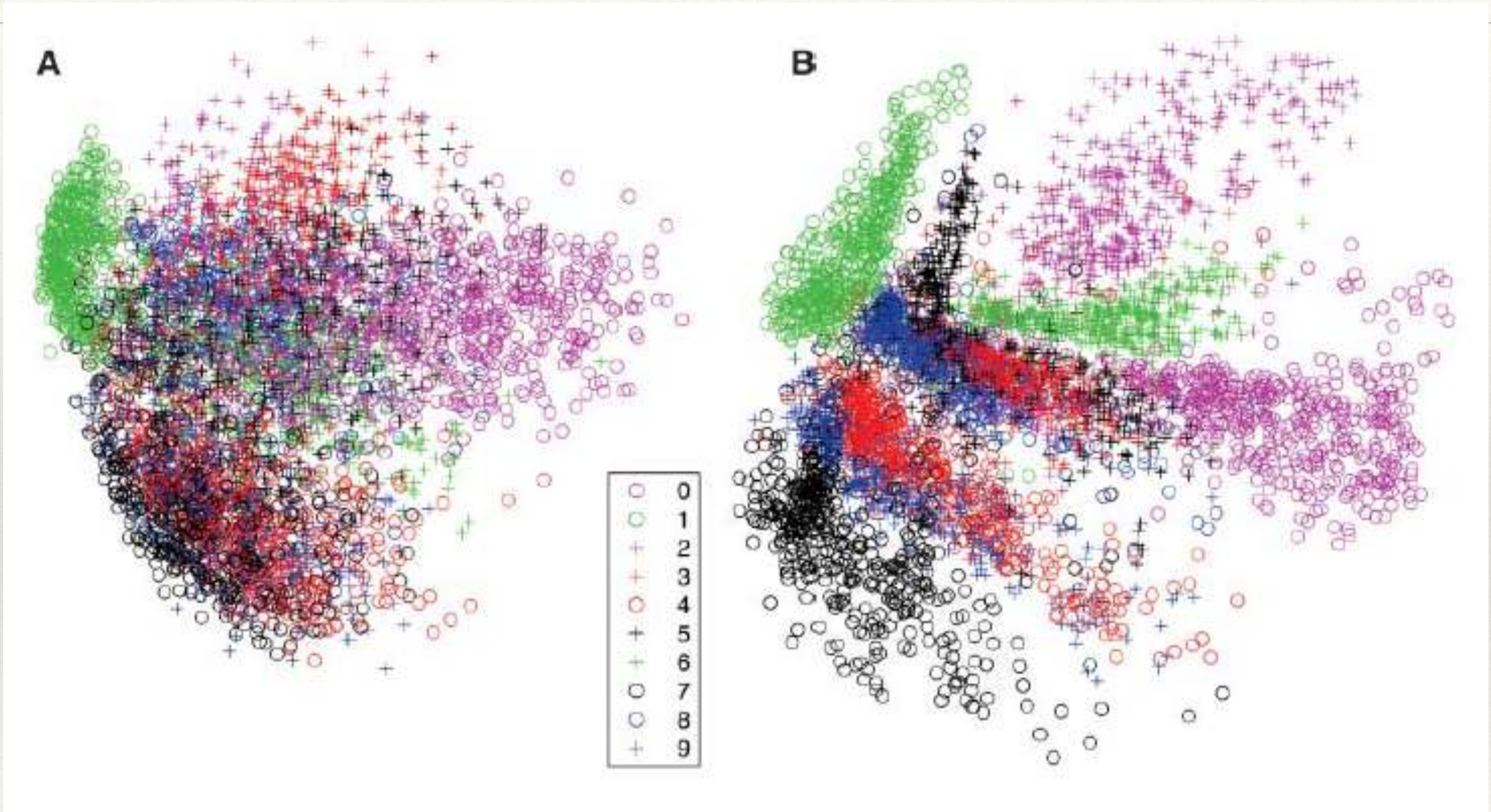
$$P(h_i = 1|\mathbf{v}) = \frac{1}{1 + e^{-(w_i^T \mathbf{v} + b_i)}}$$

RBM layers can be stacked one above the other in a **hierarchical fashion**.

Earliest application was in dimensionality reduction [Hinton, 2006].



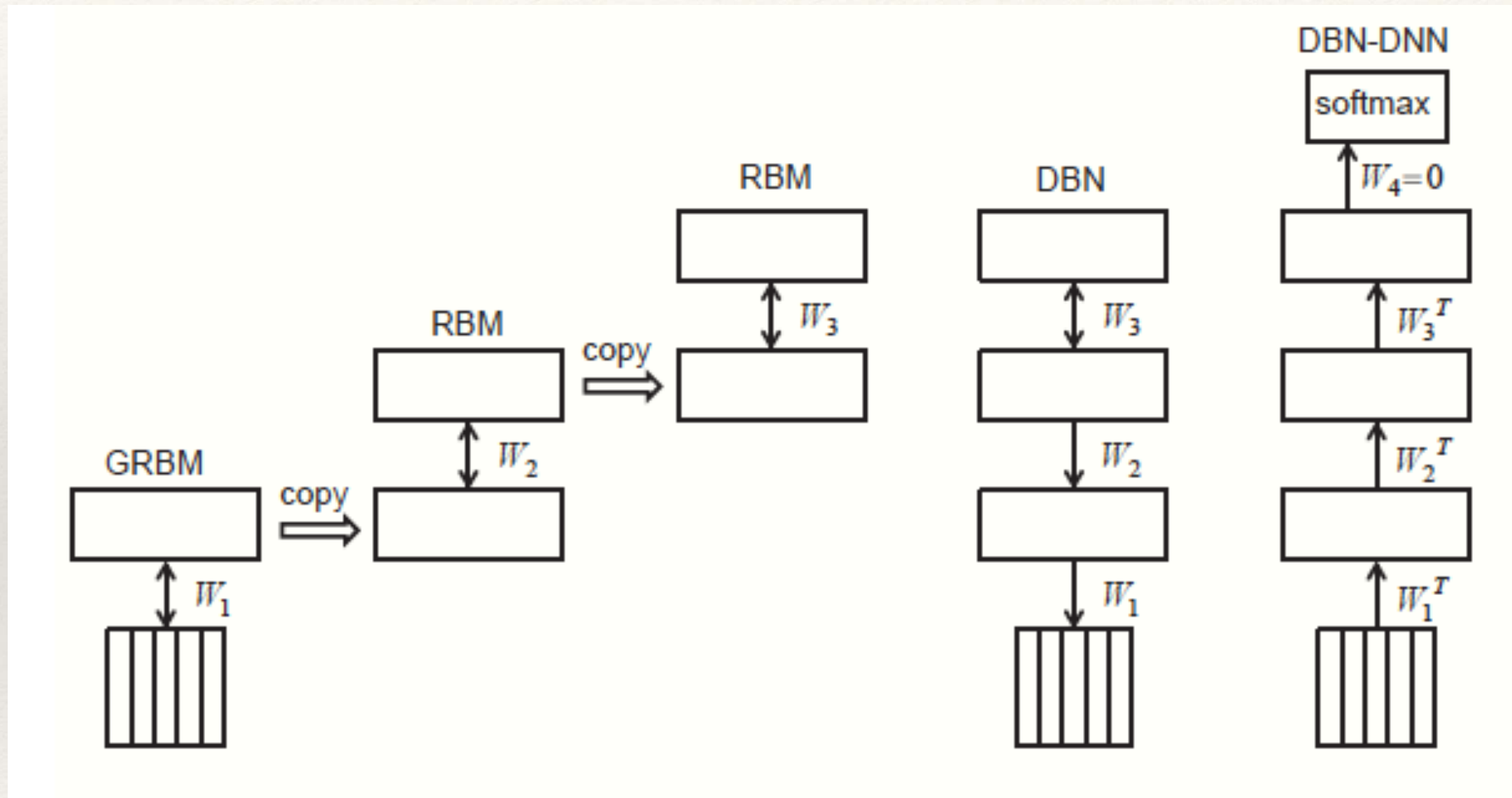
Restricted Boltzmann Machines



PCA

RBM

DBN for DNN Initialization



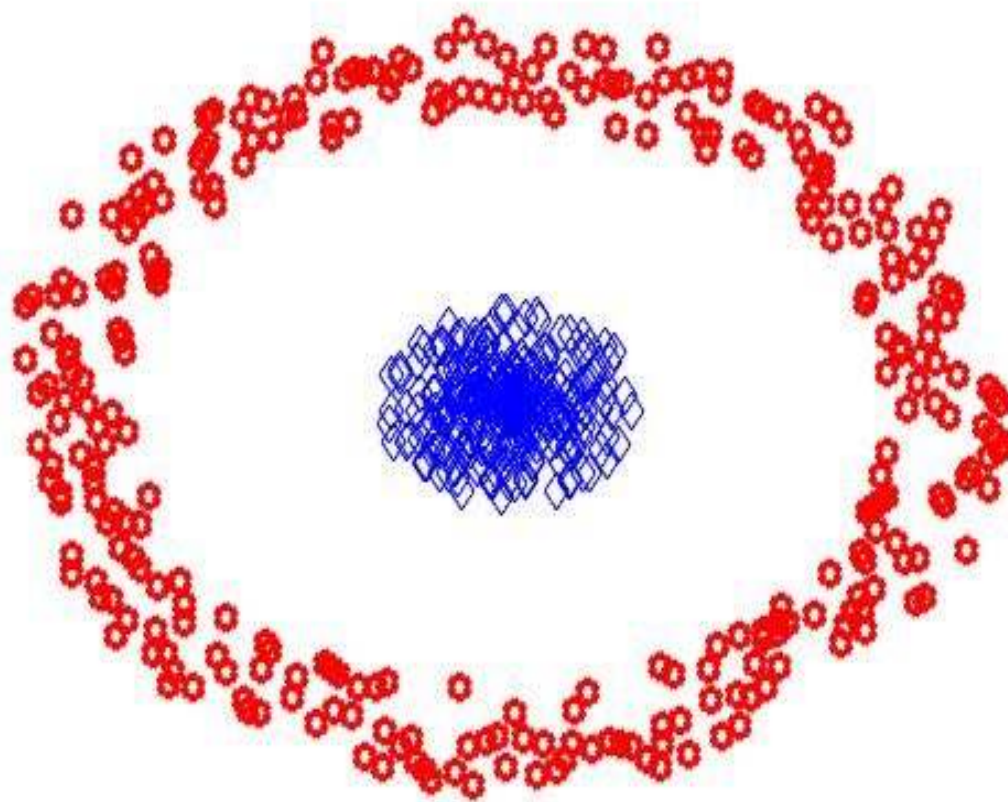
[Hinton, 2013]

Understanding Deep Networks

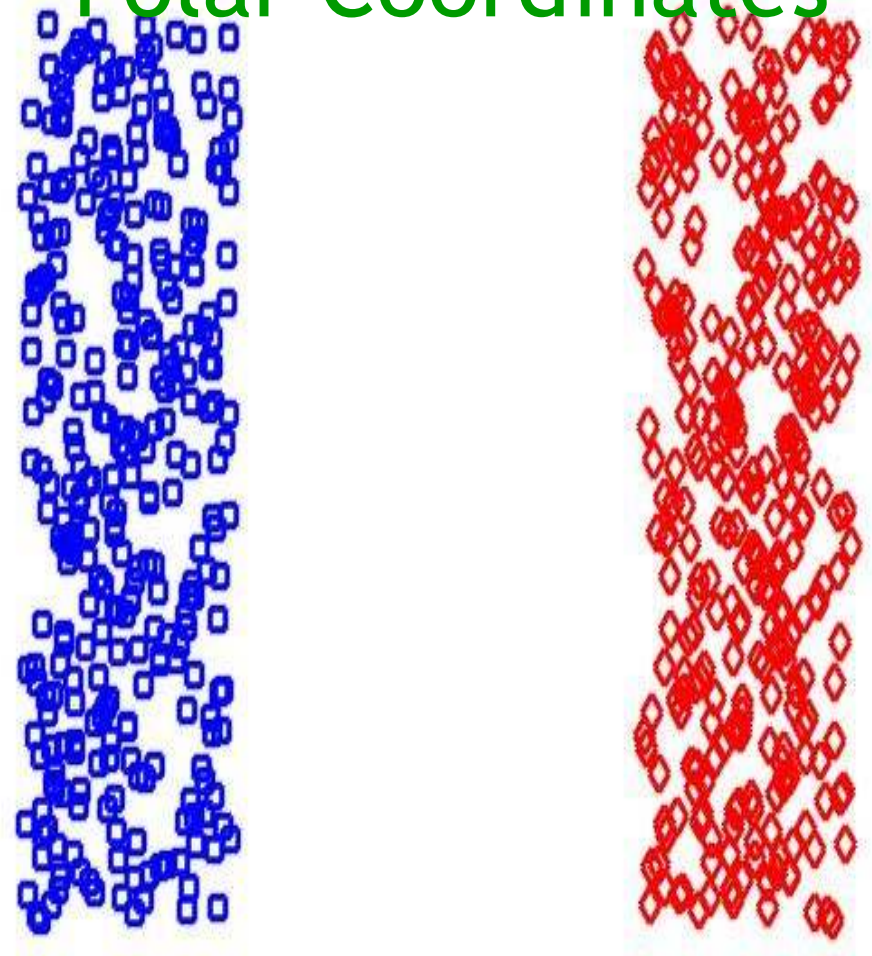
Representation Learning

- The input data representation is one of most important components of any machine learning system.

Cartesian Coordinates

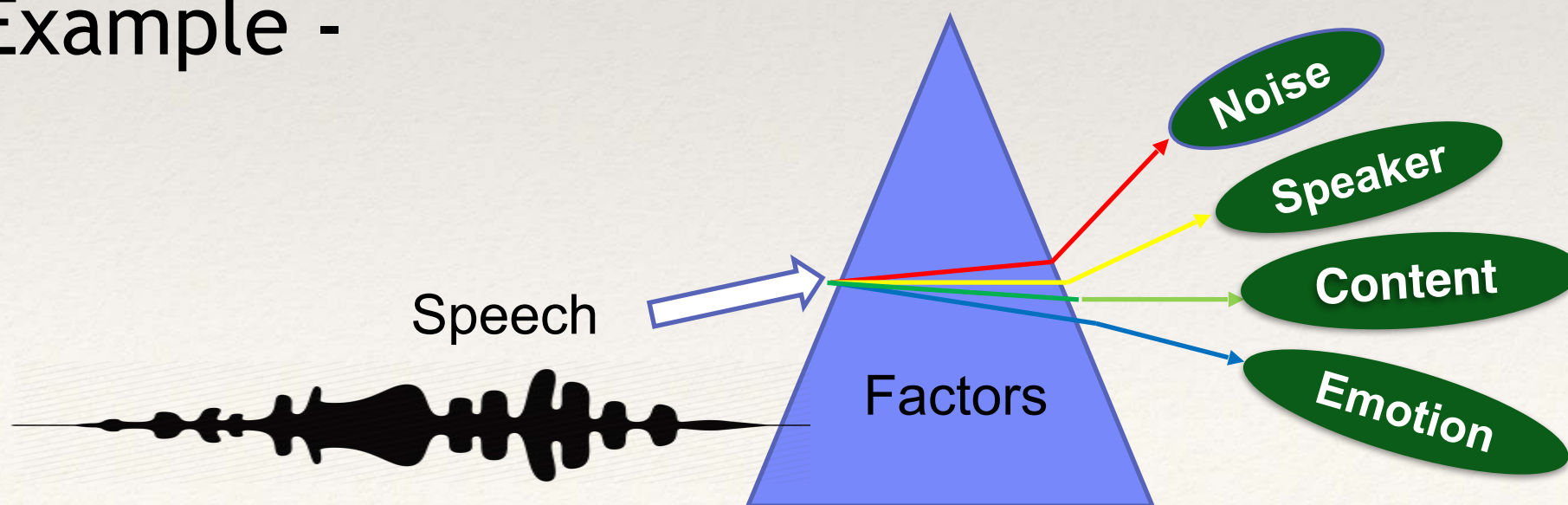


Polar Coordinates



Representation Learning

- The input data representation is one of most important components of any machine learning system.
 - Extract features that enable classification while suppressing factors which are susceptible to noise.
- Finding the right representation for real world applications substantially challenging.
 - Example -

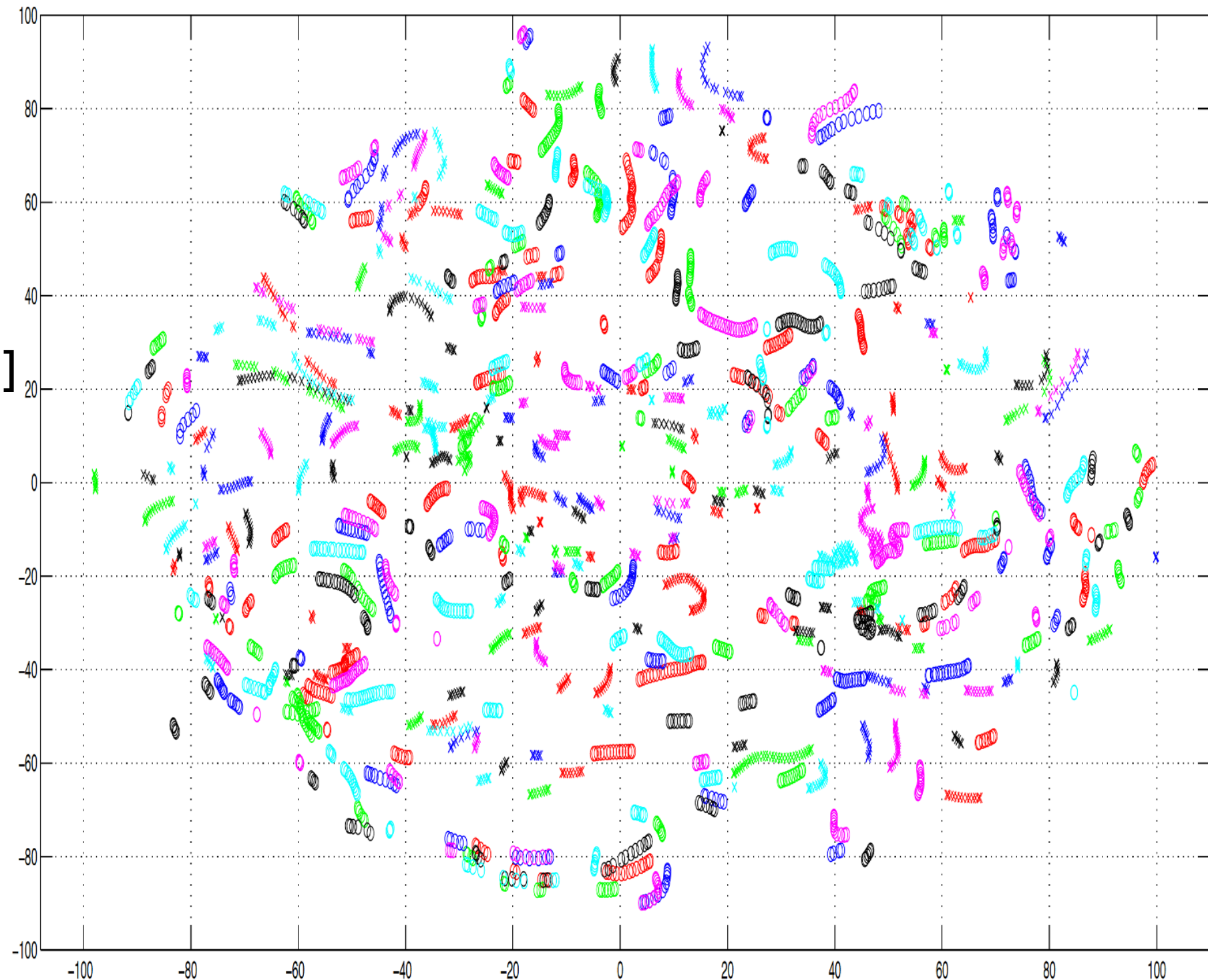


Representation Learning

- The input data representation is one of most important components of any machine learning system.
 - Extract factors that enable classification while suppressing factors which are susceptible to noise.
- Finding the right representation for real world applications substantially challenging.
 - Deep learning solution - build complex representations from simpler representations.
 - The dependencies between these hierarchical representations are refined by the target.

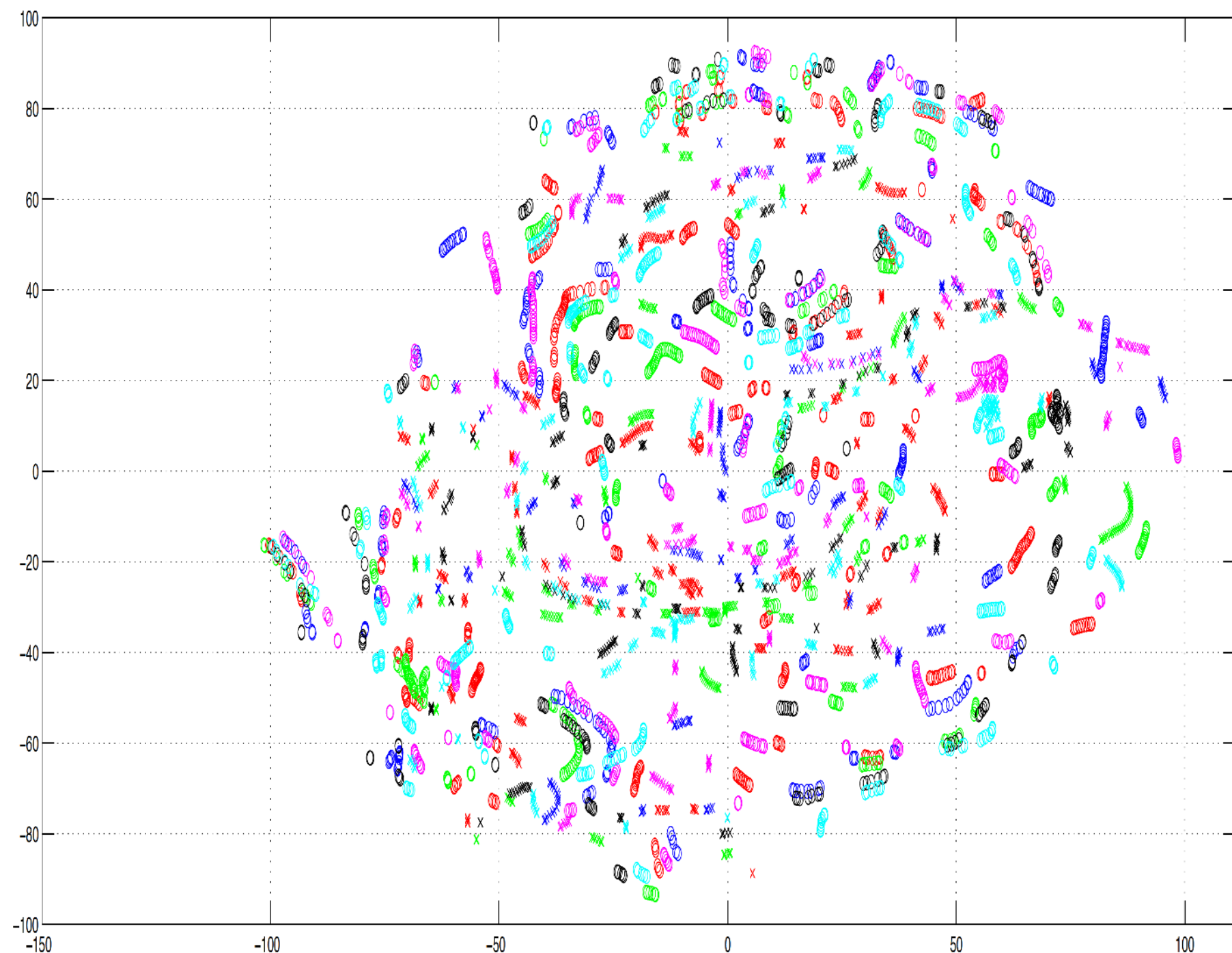
Understanding DNNs for Speech

[Abdel Rahman, 2012]



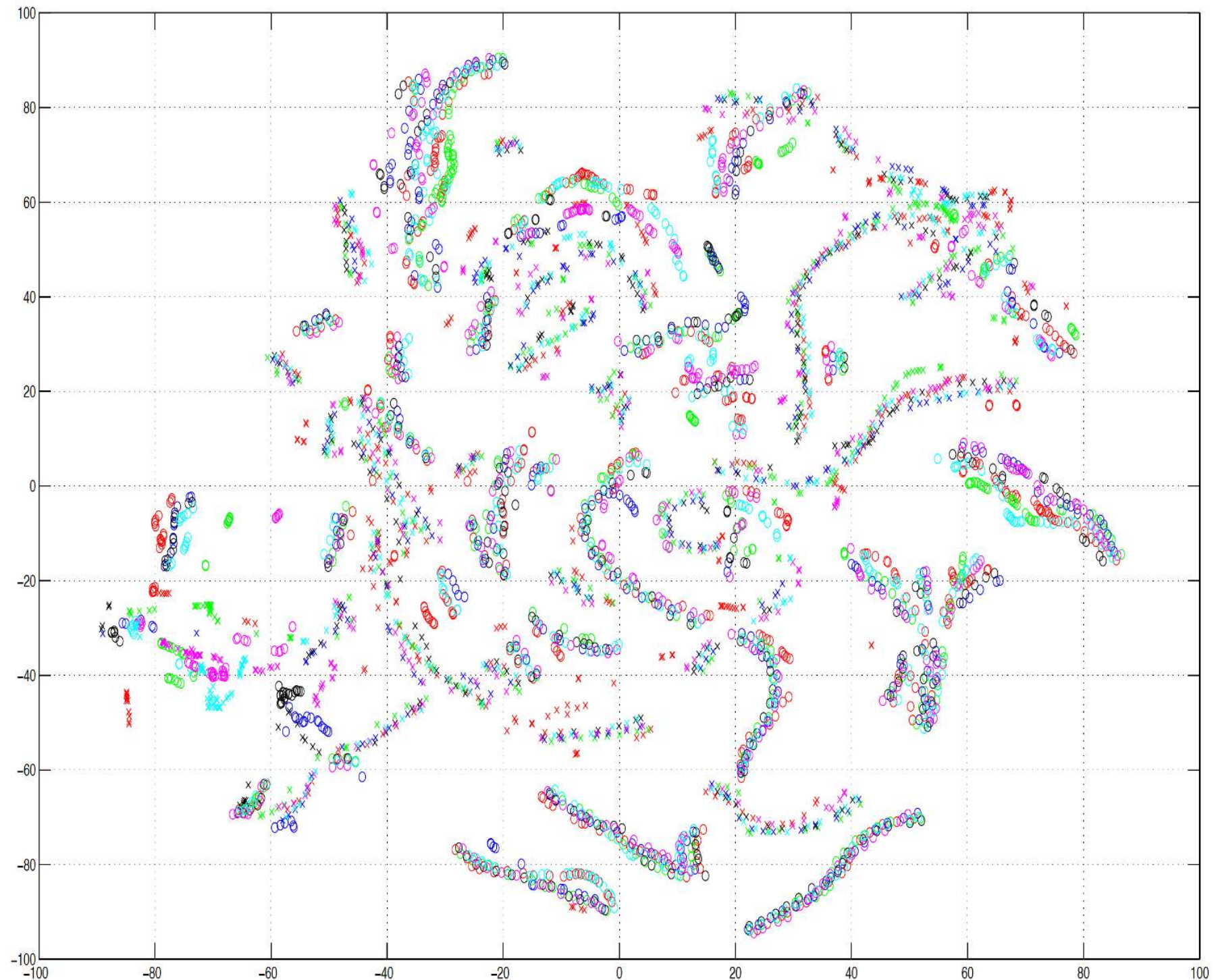
Understanding DNNs for Speech

[Abdel Rahman, 2012]

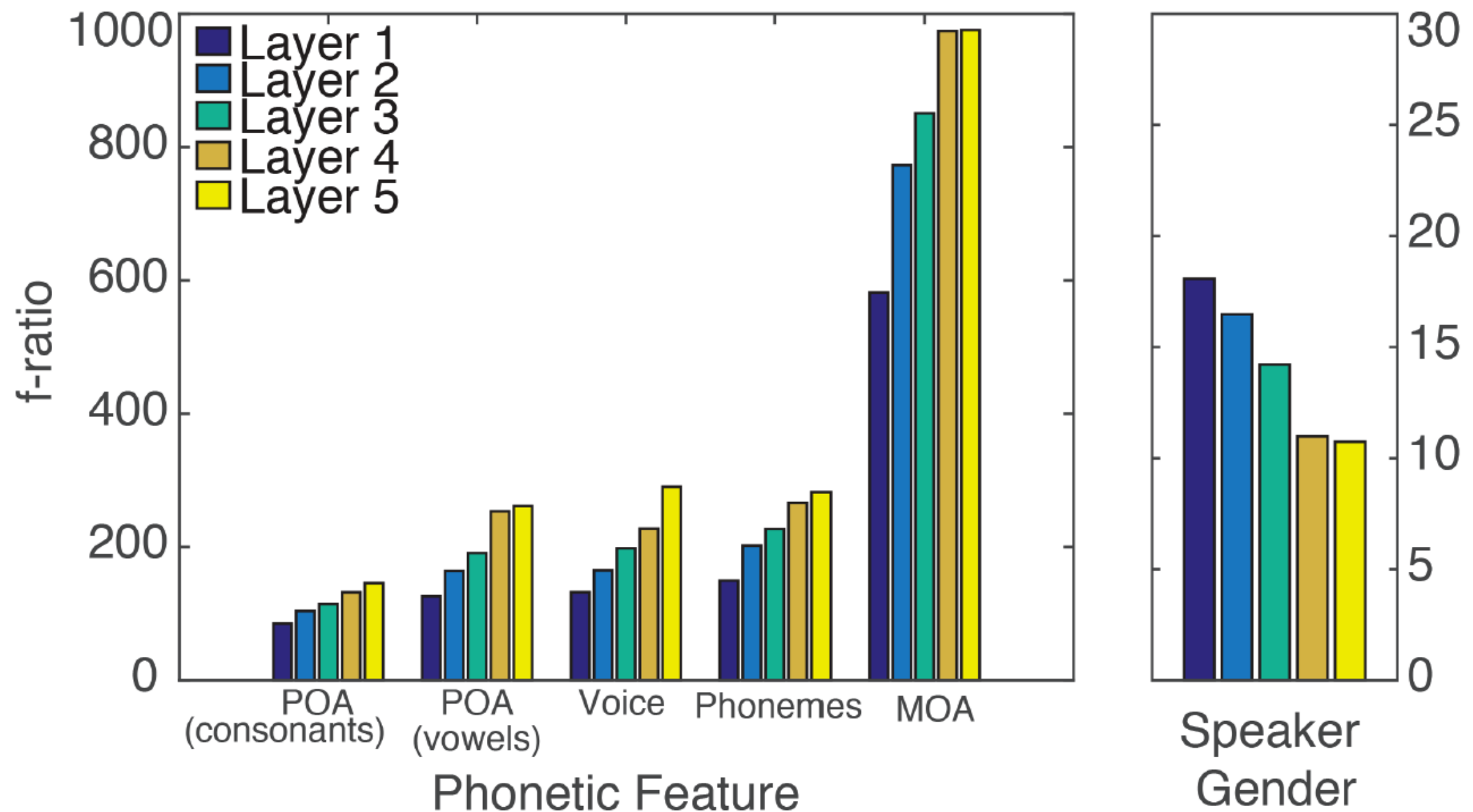


Understanding DNNs for Speech

[Abdel Rahman, 2012]



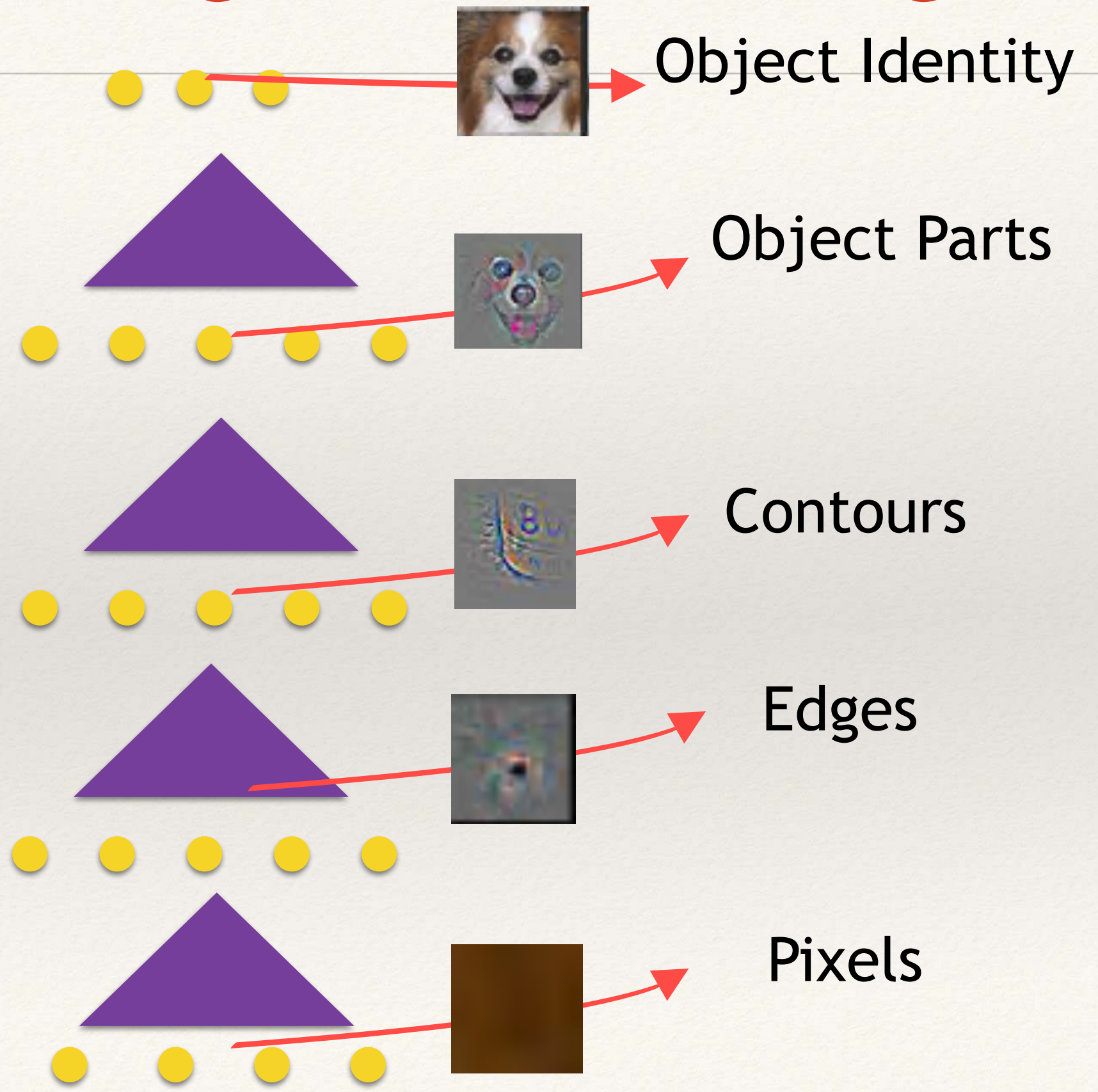
Understanding DNNs for Speech



[Nagamine, 2015]

Understanding DNNs for Image

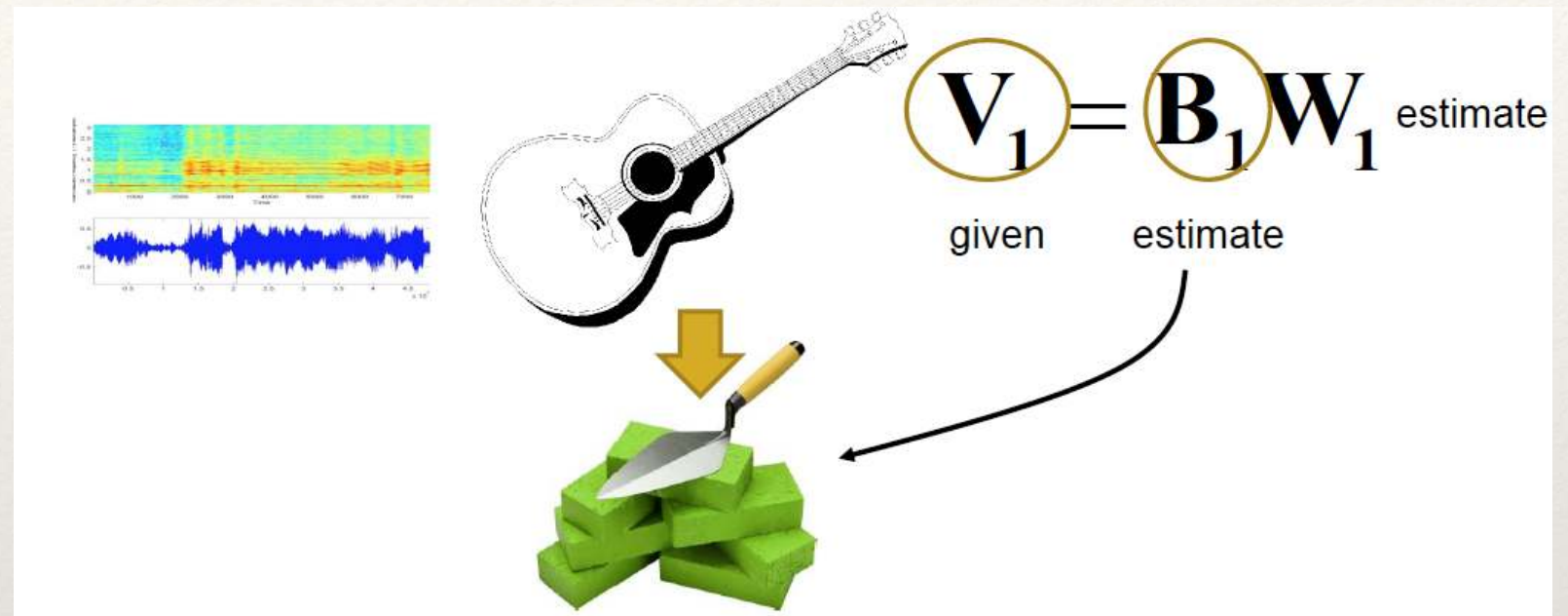
[Zeiler, 2014]



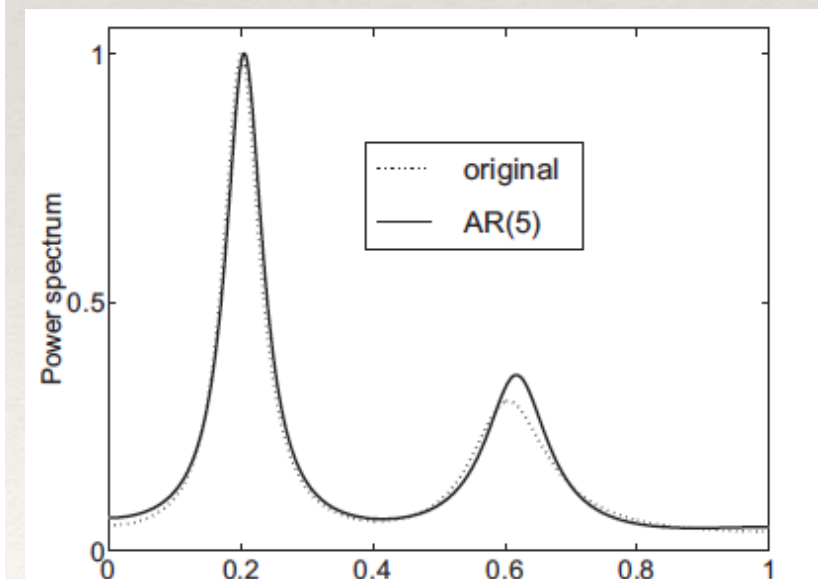
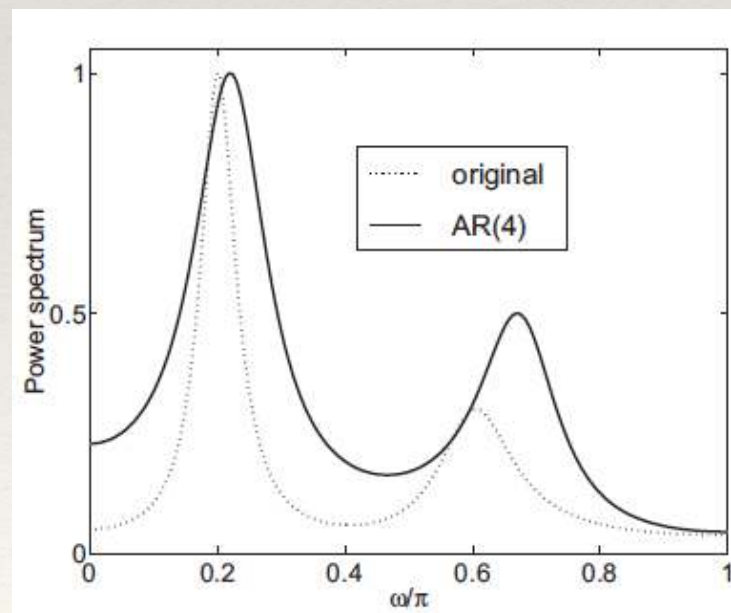
Course Summary

Introduction to Signal Processing

NMF

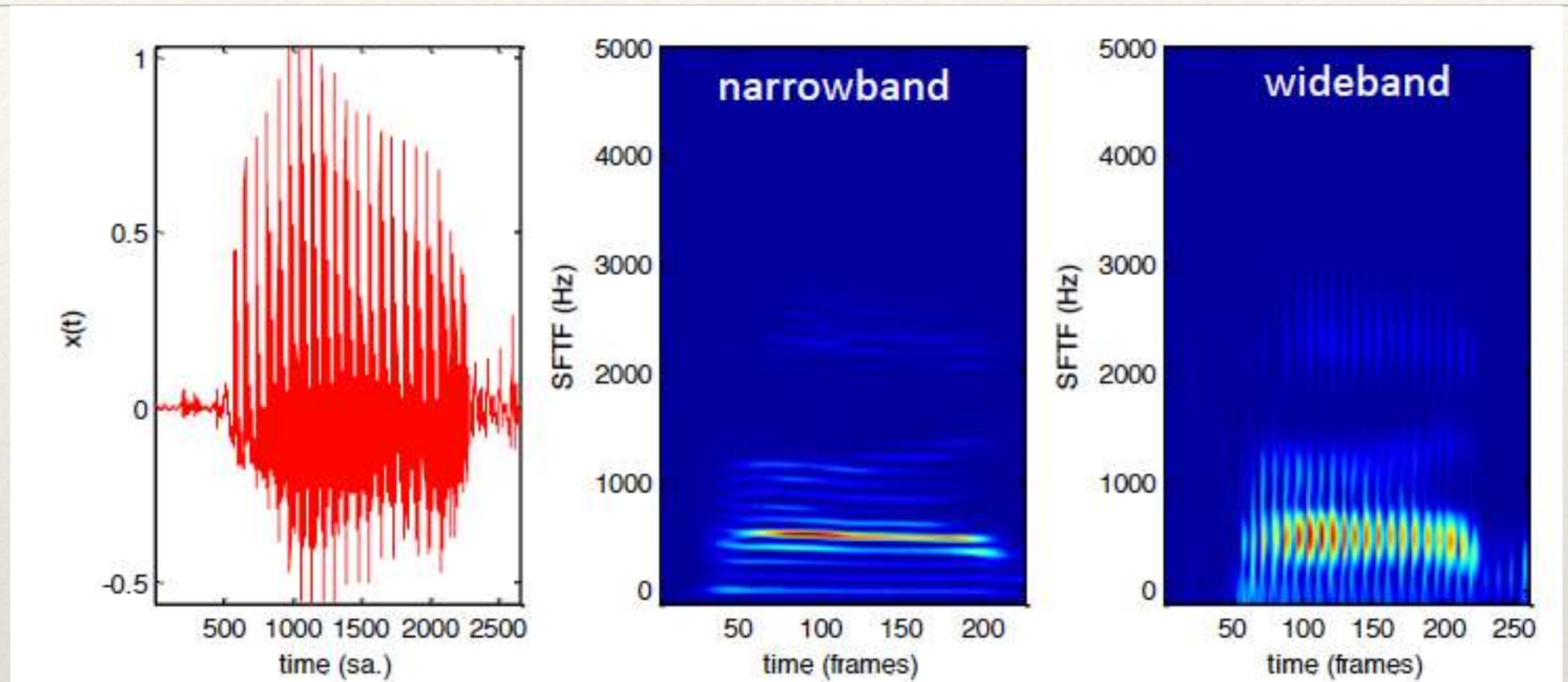


Linear Prediction



Feature Extraction

Spectrogram

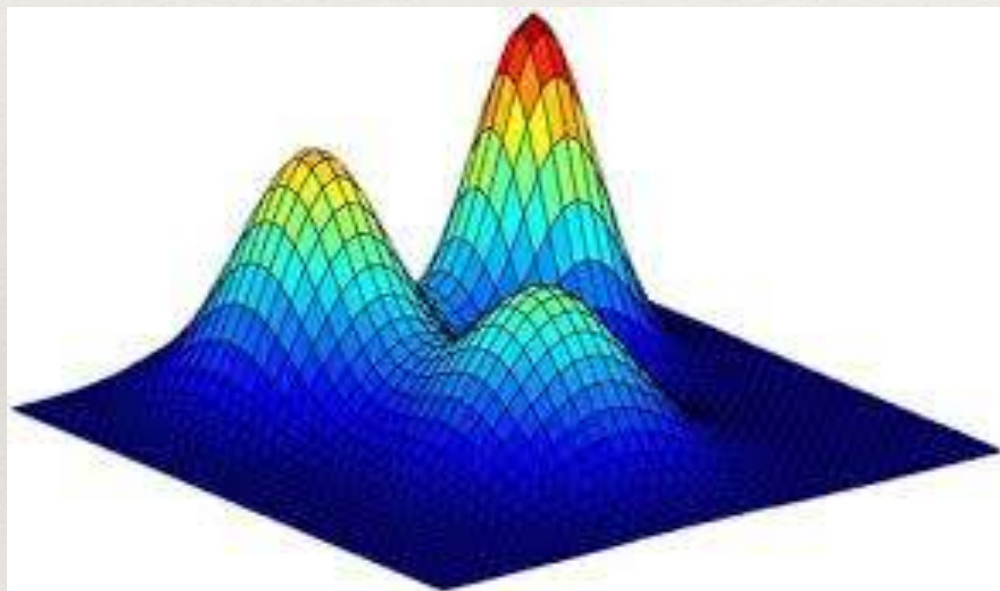


Wavelets

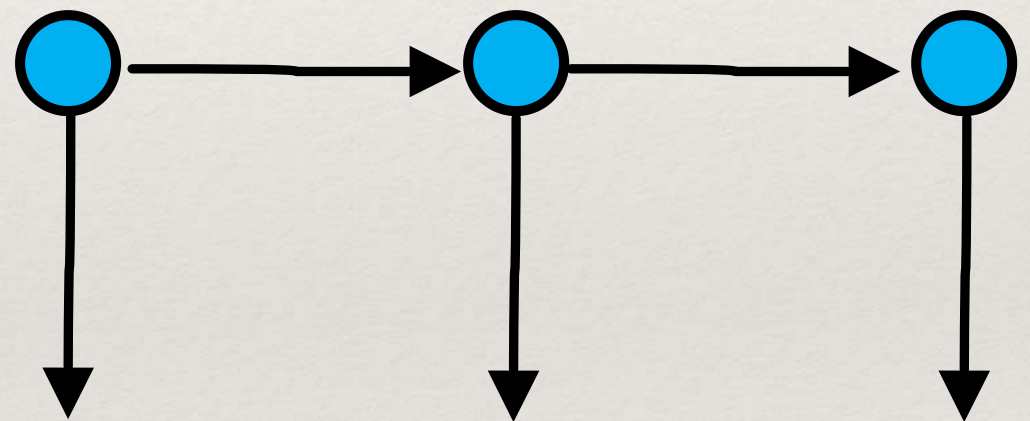


Generative Models

GMMs



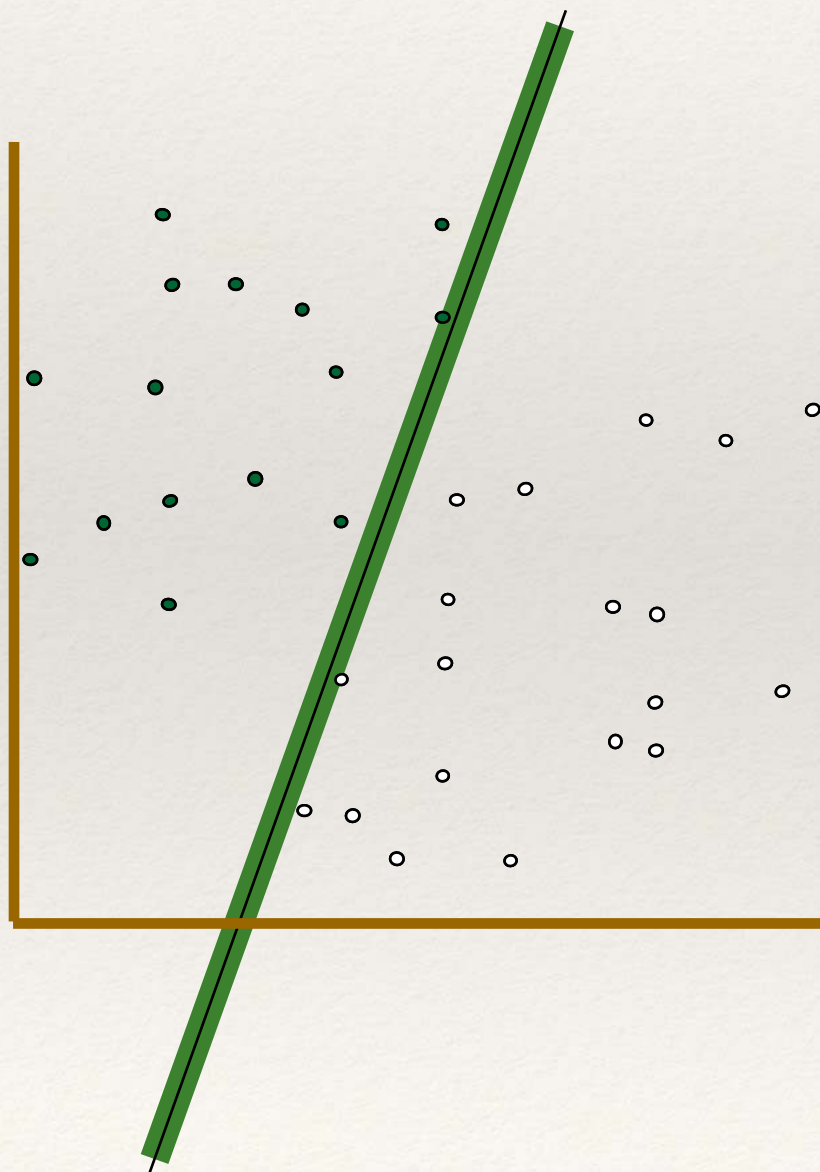
HMMs



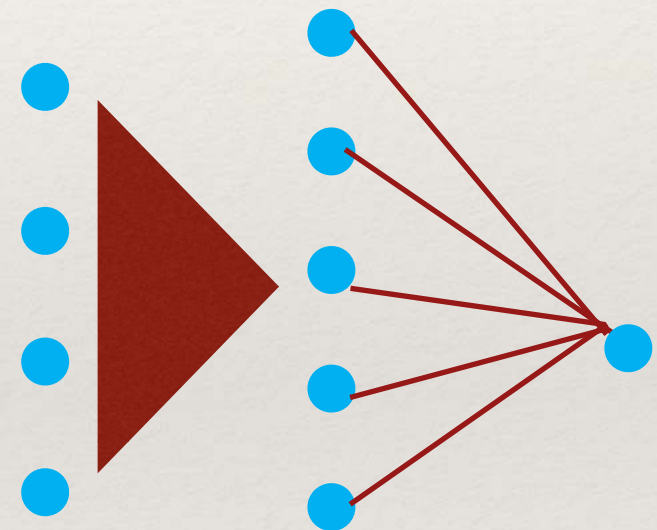
Deep Generative Models

Discriminative Models

SVMs



Neural Nets



Deep Networks

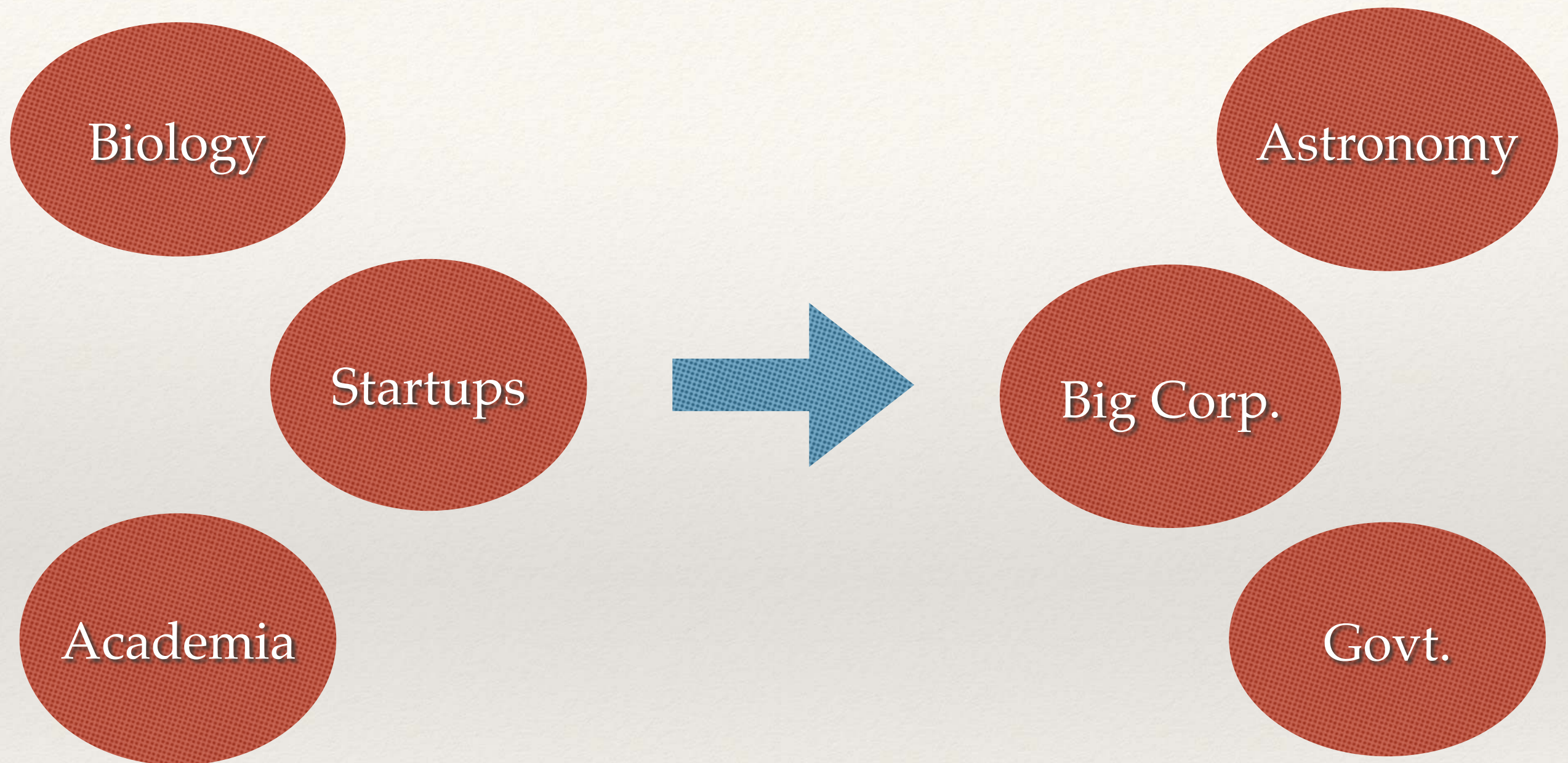
Syllabus (From Webpage)

- ❖ Introduction to real world signals
- ❖ Feature extraction and front-end signal processing
- ❖ Basics of pattern recognition, Generative modeling - Gaussian and mixture Gaussian models, hidden Markov models, factor analysis and latent variable models.
- ❖ Discriminative modeling - support vector machines, neural networks and back propagation.
- ❖ Introduction to deep learning - convolutional and recurrent networks, pre-training and practical considerations in deep learning, understanding deep networks.
- ❖ Clustering methods and decision trees. Feature selection methods.
- ❖ Applications in computer vision and speech recognition.

Syllabus (Based on Coverage)

- ❖ Introduction to real world signals
- ❖ Feature extraction and front-end signal processing
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- ❖ Deep Generative Models - RBMs and DBNs
- ❖ Applications in computer vision and speech recognition.

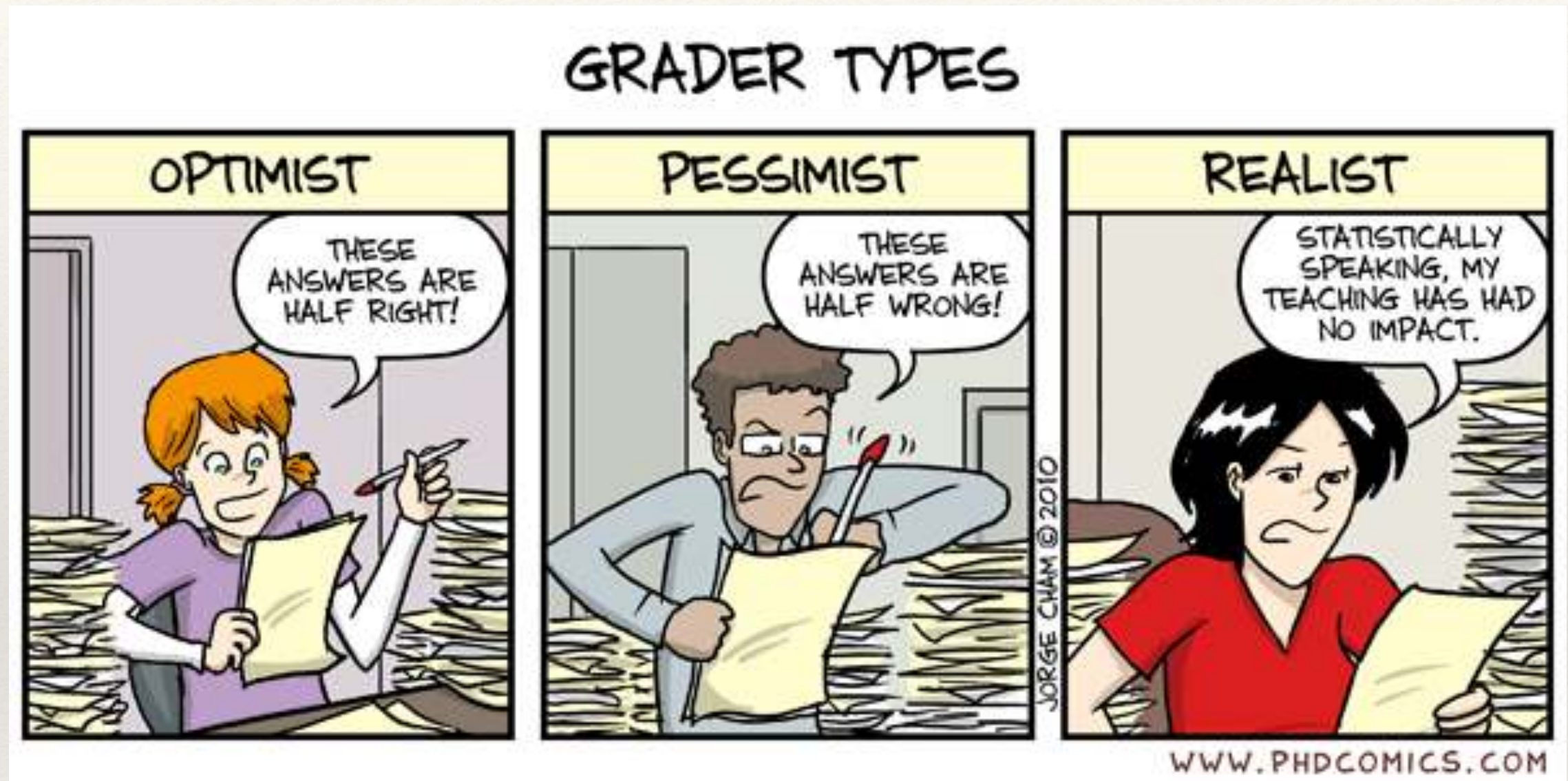
Impact



**APPLE BUYS AI STARTUP THAT
READS EMOTIONS IN FACES**

NVIDIA's Deep Learning Car Computer
Selected by Volvo on Journey Toward a
Crash-Free Future

What have we achieved ?



But I learned a lot ...